

UR5e Robot Project

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Abstract—This report presents the kinematic analysis and path planning of the UR5e 6-DOF robotic manipulator. The Denavit-Hartenberg convention was applied to derive the forward kinematics model from scratch, establishing the mathematical relationship between joint variables and the position and orientation of the end-effector relative to the base frame. The resulting forward kinematics model was validated against measurements taken directly from the physical UR5e robot using the teach pendant, with discrepancies attributed to the tool offset of the end-effector. The Jacobian matrix was then derived from the forward kinematics model to characterize velocity kinematics, and five singular configurations were identified by solving for joint configurations at which the determinant of the Jacobian vanishes. Finally, a potential field path planning algorithm was implemented for a 2-DOF planar robot, successfully generating a smooth, overshoot-free trajectory from an initial configuration to a goal configuration while avoiding a prescribed obstacle. All results were validated through MATLAB simulations using Peter Corke’s Robotics Toolbox.

Index Terms—Forward Kinematics, Jacobian Matrix, Path Planning, Potential Field, UR5e.

I. OBJECTIVE

This project focused on analyzing the UR5e robot by applying core concepts from robot synthesis. The UR5e is a 6-degree-of-freedom (DOF) robotic manipulator. The fundamental kinematics of the manipulator were examined, specifically the forward kinematics, velocity kinematics, and path planning. The Denavit-Hartenberg (D-H) convention was applied to derive the forward kinematics model from scratch, and the resulting end-effector positions were validated against measurements taken from the physical robot using the teach pendant. The Jacobian matrix was then derived from the forward kinematics model, and five singular configurations were identified. A path plan was developed using the potential field approach to move the robot from a starting configuration to a final goal while avoiding a prescribed obstacle. A MATLAB program was written from scratch to calculate this trajectory, generating the end-effector path curve and the history of joint variables across iterations. These results were used to animate the robot in Peter Corke’s Robotics Toolbox, demonstrating a smooth, fast, and overshoot-free motion that successfully avoided the obstacle.

II. BACKGROUND KNOWLEDGE

A. Denavit-Hartenberg (D-H) Convention

The Denavit-Hartenberg convention is a standardized method for modeling the geometry of robotic manipulators. It served as a basis for the forward kinematics of the robotic manipulator, providing a mathematical approach to calculating an end-effector’s position and orientation relative to the base frame. A robotic manipulator must meet the D-H conditions,

such that each link had a frame assigned to it, and each frame’s orientation must meet particular axis intersections between neighboring frames.

Forward kinematics used the known joint variables to determine the position and orientation of the end-effector relative to the base [1]. Fig. 1 shows the UR5e joints and kinematic dimensions (mm). The Denavit-Hartenberg (DH) convention is used to calculate the forward kinematics equations. This procedure assigned a relative coordinate frame to each link.

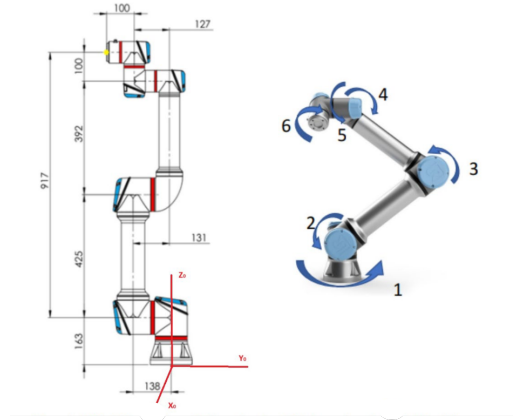


Fig. 1. UR5e joints and kinematic dimensions (mm). Joints are shown in red on the left and are marked with arrows on the right.

B. The Jacobian Matrix

The Jacobian matrix,

$$(J(\mathbf{q}))$$

represented a mathematical relationship between the velocities of each joint and the resulting motion of the end-effector. In particular, it mapped the joint velocity vector into the translational and angular velocity of the end-effector relative to the base frame. This provided a direct connection between joint space (joint angles and joint rates) and task space (position and orientation of the robot’s end-effector).

The Jacobian was constructed by combining the contributions of each joint in the kinematic chain, where each column corresponded to a specific joint and captured how that joint influenced both linear and angular motion. The upper portion of the matrix described the effect of joint motion on the end-effector’s linear velocity, while the lower portion described its effect on angular velocity. In this way, the Jacobian encoded how motion propagated through each link of the robot, accumulating to produce the final end-effector velocity.

Beyond velocity analysis, the Jacobian is also essential in force and torque relationships. The transpose of the Jacobian maps forces and moments applied at the end-effector back into equivalent joint torques. This allows external interactions—such as pushing, grasping, or contacting objects—to be translated into the required actuator efforts at each joint, to make the Jacobian a key component in robot dynamics, control, and interaction tasks.

C. Path Planning

Path planning aimed to create a collision-free path to move the robot from its initial configuration to its final configuration. The potential field approach was used, which operates on the principle of two artificial forces acting on each joint origin. An attractive force pulled each joint origin toward its desired goal position, while a repulsive force deflected each joint origin away from any obstacle within a prescribed radius of influence. The superposition of these two forces at each joint origin was converted into joint torques via the Jacobian transpose method, and the resulting torques were used to iteratively update the joint configuration until the end-effector converged to the goal.

D. Singularities

A singularity occurred when the Jacobian matrix $\mathbf{J}$ of a robotic manipulator became singular at a specific configuration. At such a configuration, it is mathematically impossible to accurately predict the realistic movement of the robot arm. This causes the manipulator to lose the ability to move in certain directions, making those directions of motion unattainable.

Singularities present two critical implications for manipulator behavior. First, bounded end-effector velocities may correspond to unbounded joint velocities, meaning that although the end-effector velocity remains finite, the required joint velocities approach infinity. Second, bounded joint torques may correspond to unbounded end-effector forces, where finite joint torques result in infinite end-effector forces. These conditions make singularities particularly dangerous during robot operation.

E. Universal Robots e-Series UR5e: Integration of the Robot Operating System

The Universal Robots UR5e (e-Series) is a 6-DOF collaborative robot arm that is widely used in research and industry for automation tasks such as pick-and-place, assembly, and manipulation. It integrates with the Robot Operating System (ROS/ROS2) through official and community-supported drivers. With ROS integration, the UR5e can be controlled in real time using frameworks such as MoveIt, allowing users to perform motion planning, trajectory execution, and Cartesian control of the end-effector in simulation or on physical hardware through standardized ROS 2 control pipelines. The robot includes built-in force/torque sensing and high repeatability, making it suitable for compliant motion, force-based control, and human-robot interaction tasks.

F. Peter Corke's Robotics Toolbox

Peter Corke's Robotics Toolbox demonstrated the functionality of arm-type robotics using simulations. Inside MATLAB, a serial-link manipulator can be created using a D-H Convention, which can be used to plot the arm's final configuration given specific theta values of each joint. This approach highlights the algorithms and calculations required to calculate these configurations.

III. CALCULATION AND PROCEDURES

A. Denavit-Hartenberg (D-H) Convention

In order to meet D-H conditions, two frames with axes $o_{i-1}, x_{i-1}, y_{i-1}, z_{i-1}$ and o_i, x_i, y_i, z_i must meet the following two conditions:

- 1) The axis x_i is perpendicular to the axis z_{i-1} .
- 2) The axis x_i must intersect the axis z_{i-1} .

These frames were defined by aligning each z_{i-1} axis with the actuation axis of joint J_i . The x_i axes were then assigned according to the D-H convention, ensuring consistency with the geometric relationships between consecutive links.

There exist 4 parameters, a, d, θ, α , such that:

- a_i : the distance between z_{i-1} and z_i along x_i .
- d_i : the distance from the origin o_{i-1} to the intersection of axis x_i with z_{i-1} along axis z_{i-1} .
- α_i : the angle between the z_{i-1} and z_i axes, measured about the x_i axis.
- θ_i : the angle between the x_{i-1} and x_i axes, measured about the z_{i-1} axis.

If these parameters exist, then the two frames meet the D-H conditions. The Homogeneous matrix H , are the forward kinematics equations that relate the position and orientation of one frame to another, can be found via:

$$\mathbf{H}_i^{i-1} = \mathbf{R}_z(\theta_i) \cdot \mathbf{Trans}_z(d_i) \cdot \mathbf{Trans}_x(a_i) \cdot \mathbf{Rot}_x(\alpha_i) \quad (1)$$

The basic transformation matrices are defined as:

$$\mathbf{T}_x(a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$\mathbf{T}_z(d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$$\mathbf{R}_z(\theta_i) = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$\mathbf{R}_x(\alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha_i & -\sin \alpha_i & 0 \\ 0 & \sin \alpha_i & \cos \alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

The completed transformation from the base frame to the end-effector is represented by the matrix $\mathbf{H}_6^0$:

$$\mathbf{H}_6^0 = \mathbf{H}_1^0 \cdot \mathbf{H}_2^1 \cdot \mathbf{H}_3^2 \cdot \mathbf{H}_4^3 \cdot \mathbf{H}_5^4 \cdot \mathbf{H}_6^5 \quad (6)$$

TABLE I
DH PARAMETERS FOR THE ROBOTIC MANIPULATOR

Frame Transformations	a_i (mm)	d_i (mm)	α_i (rad)	θ_i (rad)
$0 \rightarrow 1$	0	163	$-\pi/2$	$\pi + \theta_1$
$1 \rightarrow 2$	425	138	π	θ_2
$2 \rightarrow 3$	392	131	π	$-\theta_3$
$3 \rightarrow 4$	0	127	$-\pi/2$	θ_4
$4 \rightarrow 5$	0	100	$-\pi/2$	$\pi + \theta_5$
$5 \rightarrow 6$	0	247	0	θ_6

Using DH Table I, we plugged in the parameter values in equation (1) for all 6 frame transformations, then plugged the results into the H_6^0 matrix in equation (6).

B. The Jacobian Matrix

For the UR5e, the Jacobian related the velocities of the six revolute joints, represented by the variable $\dot{\mathbf{q}}$, to the linear and angular velocities of the end effector, represented by $\xi = [\mathbf{v}, \omega]^T$. The relationship is defined as:

$$\xi = J(\mathbf{q})\dot{\mathbf{q}} \quad (7)$$

Within the Jacobian, the translational component J_{vi} and angular component $J_{\omega i}$ for joint i are:

$$J_{vi} = \begin{cases} z_{i-1}^0 \times (O_n^0 - O_{i-1}^0) & \text{(revolute)} \\ z_{i-1}^0 & \text{(prismatic)} \end{cases} \quad (8)$$

$$J_{\omega i} = \begin{cases} z_{i-1}^0 & \text{(revolute)} \\ \mathbf{0} & \text{(prismatic)} \end{cases} \quad (9)$$

Where J_v and J_ω are made of:

$$J_v = \begin{bmatrix} J_{xv} \\ J_{yv} \\ J_{zv} \end{bmatrix}, \quad \text{and} \quad J_\omega = \begin{bmatrix} J_{x\omega} \\ J_{y\omega} \\ J_{z\omega} \end{bmatrix} \quad (10)$$

The complete Jacobian matrix is a $6 \times i$ matrix where i is the number of joints in the robotic arm. Using the components above, the Jacobian matrix is arranged as:

$$J_{6 \times i} = \begin{bmatrix} J_{vi} \\ J_{\omega i} \end{bmatrix} \quad (11)$$

The complete Jacobian is then multiplied by the joint velocity vector $\dot{\mathbf{q}}$, which contains the angular velocities of each revolute joint. This product yields the end-effector velocity vector, expressed as

$$\xi = J(\mathbf{q})\dot{\mathbf{q}}, \quad (12)$$

where $\xi = [\mathbf{v}; \omega]$ contains the translational and angular velocity of the end-effector relative to the base frame.

C. Path Planning

Path planning is performed using an artificially implemented potential field method, where attractive forces draw the end-effector toward the goal configuration and repulsive forces deflect the manipulator away from obstacles. Denavit–Hartenberg frames are first attached to each link to establish consistent coordinate systems throughout the kinematic chain.

The initial and final joint configurations are defined as q_s and q_f respectively, from which the Cartesian positions of each joint origin O_i are determined at both configurations as

$$q_s = \begin{bmatrix} \theta_1^s \\ \theta_2^s \end{bmatrix}, \quad q_f = \begin{bmatrix} \theta_1^f \\ \theta_2^f \end{bmatrix}. \quad (13)$$

At each iteration, the attractive force acting on joint origin O_i is computed as the scaled displacement between its current and goal positions,

$$F_{\text{att},i}(q) = -\xi_i [O_i(q) - O_i(q_f)], \quad (14)$$

where $\xi_i > 0$ is the attractive gain for the i -th point. The repulsive force is nonzero only when O_i lies within the radius of influence ρ_0 of an obstacle located at coordinate point b , and is given by

$$F_{\text{rep},i}(q) = \eta_i \left[\frac{1}{\rho(O_i)} - \frac{1}{\rho_0} \right] \frac{1}{\rho^2(O_i)} \cdot \frac{O_i(q) - b}{|O_i(q) - b|}, \quad (15)$$

where $\eta_i > 0$ is the repulsive gain and $\rho(O_i) = |O_i(q) - b|$ is the Euclidean distance from O_i to the obstacle. The total force at each point is then the superposition of both contributions,

$$F_i(q) = F_{\text{att},i}(q) + F_{\text{rep},i}(q). \quad (16)$$

These Cartesian forces are mapped into joint torques using the Jacobian transpose method, accumulating contributions from all relevant joint origins as

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \sum_i J_{O_i}^T(q) \cdot F_i(q). \quad (17)$$

The Jacobian matrix above is calculated differently from the Jacobian matrix previously defined. This Jacobian matrix and its preceding components were specially calculated with the assumption that a given joint acts as the end effector.

$$J_{v1} = z_{i-1}^0 \times (O_n^0 - O_{i-1}^0) \quad (18)$$

where K_1 and K_2 are positive scalars that act as gain values that govern the step size. The robot is then considered to be moved to the updated configuration, where the new values of theta act as the angles of each joint. The process is repeated from the force computation stage until the end-effector approaches the desired goal configuration q_f or a maximum number of iterations is reached.

D. Singularities

For an $m \times n$ matrix, a singularity occurs when the rank of the Jacobian matrix, r , is less than the number of rows (m) and/or the number of columns (n).

The singularities of the robot were identified by the values of the configuration where the determinant of the Jacobian equals zero. This ensures that the rank of the matrix is always less than the number of columns or rows in an $m \times n$ matrix.

The kinematic singularities of the robotic system are identified by the configurations in which the determinant of the Jacobian matrix $J(\theta) \in \mathbb{R}^{m \times n}$ vanishes.

Physically, these configurations commonly manifest at the workspace boundaries, occurring when the manipulator is either fully extended at its maximum reach or folded back upon itself. In such states, the manipulator loses one or more degrees of freedom, meaning no combination of joint velocities can produce motion in specific Cartesian directions, which is formally expressed as

$$\text{rank}(J(\theta)) < \min(m, n). \quad (19)$$

This rank deficiency necessitates careful trajectory planning to avoid actuator saturation and numerical instability within the control system. The singular configurations are therefore identified by solving

$$\det(J(\theta)) = 0. \quad (20)$$

IV. RESULTS

The positions of the end-effector relative to the base frame are predicted in Table II. These values were calculated from matrix $\mathbf{H}_6^0$ using parameters defined in (1)–(5).

TABLE II
END-EFFECTOR POSITION PREDICTED BY FORWARD KINEMATICS

Joint Angles (deg)						End-effector Position (mm)		
θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	x	y	z
0	-90	0	-90	0	0	0	-234	1080
18	-43	7.8	-60	28	40	-622.23	-435.91	734.63
61	-125	74	-11	77	80	70.473	-195.66	854.87
79	-130	-23	8	64	108	296.44	593.06	800
-36	-136	77	-90	70	153	8.6259	-214.18	928.35

The position displayed on the teach pendant screen was logged in Table III.

The predicted positions in Table II differ from Table III because the former does not account for the tool length. The additional length of the end effector tool was found to be 174.548 mm. Table IV shows the corrected predictions considering this tool length.

The Jacobian matrix was then derived from the forward kinematics results. The origin vectors O_i^0 and axis unit vectors z_i^0 were extracted from the H matrices evaluated at the home

TABLE III
END-EFFECTOR POSITION MEASURED VIA TEACH PENDANT

Joint Angles (deg)						End-effector Position (mm)		
θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	x	y	z
0	-90	0	-90	0	0	0.06	-408.61	1079.63
18	-43	7.8	-60	28	40	-567.25	-580.47	815.77
61	-125	74	-11	77	80	66.25	-284.50	1004.76
79	-130	-23	8	64	108	396.47	704.85	889.70
-36	-136	77	-90	70	153	87.10	-345.23	1012.51

TABLE IV
PREDICTED POSITION CONSIDERING TOOL OFFSET

Joint Angles (deg)						Position (mm)		
θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	x	y	z
0	-90	0	-90	0	0	0	-408.55	1080
18	-43	7.8	-60	28	40	-567.54	-580.19	816.24
61	-125	74	-11	77	80	66.11	-284.53	1005
79	-130	-23	8	64	108	396.08	704.61	889.99
-36	-136	77	-90	70	153	87.28	-345.11	1012.8

configuration ($\theta_1 = \dots = \theta_6 = 0^\circ$, $\theta_2 = -90^\circ$, $\theta_4 = -90^\circ$), as shown below:

$$\begin{aligned} O_0^0 &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, & O_1^0 &= \begin{bmatrix} 0 \\ 0 \\ 163 \end{bmatrix}, & O_2^0 &= \begin{bmatrix} 0 \\ -138 \\ 588 \end{bmatrix} \\ O_3^0 &= \begin{bmatrix} 0 \\ -7 \\ 980 \end{bmatrix}, & O_4^0 &= \begin{bmatrix} 0 \\ -134 \\ 980 \end{bmatrix}, & O_5^0 &= \begin{bmatrix} 0 \\ -134 \\ 1080 \end{bmatrix} \end{aligned} \quad (21)$$

The axis vectors are:

$$\begin{aligned} z_0^0 &= [0, 0, 1]^T, & z_1^0 &= [0, -1, 0]^T, & z_2^0 &= [0, 1, 0]^T \\ z_3^0 &= [0, -1, 0]^T, & z_4^0 &= [0, 0, 1]^T, & z_5^0 &= [0, -1, 0]^T \end{aligned} \quad (22)$$

The linear velocity columns J_{v_i} and angular velocity columns J_{ω_i} are calculated as follows:

$$\begin{aligned} J_{v_1} &= \begin{bmatrix} 234 \\ 0 \\ 0 \end{bmatrix}, & J_{v_2} &= \begin{bmatrix} -917 \\ 0 \\ 0 \end{bmatrix}, & J_{v_3} &= \begin{bmatrix} 492 \\ 0 \\ 0 \end{bmatrix} \\ J_{v_4} &= \begin{bmatrix} -100 \\ 0 \\ 0 \end{bmatrix}, & J_{v_5} &= \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}, & J_{v_6} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned} \quad (23)$$

The angular velocity components are:

$$\begin{aligned} J_{\omega_1} &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, & J_{\omega_2} &= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, & J_{\omega_3} &= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ J_{\omega_4} &= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, & J_{\omega_5} &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, & J_{\omega_6} &= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \end{aligned} \quad (24)$$

The resulting full Jacobian matrix J is given by:

$$J = \begin{bmatrix} 234.0 & -917.0 & 492.0 & -100.0 & 100.0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1.0 & 1.0 & -1.0 & 0 & -1.0 \\ 1 & 0 & 0 & 0 & 1.0 & 0 \end{bmatrix} \quad (25)$$

The joint configurations $\theta = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6]^T$ and the corresponding Jacobian matrices $J(\theta)$ identifying singular states are presented below. Values are rounded to two decimal places for column compatibility.

Config 1: $\theta = [0, 2.74, 0, 0, 0, 0]^T$

$$J_1 = \begin{bmatrix} 234 & 717 & -292 & -100 & 100 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Config 2: $\theta = [0, -0.40, 1.0, 0, 0, 0]^T$

$$J_2 = \begin{bmatrix} 234.0 & 632.45 & -240.83 & -82.48 & 82.48 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -165.11 & -56.55 & 56.55 & 0 \\ 0 & 0 & 0 & 0 & -0.57 & 0 \\ 0 & -1 & 1 & -1 & 0 & -1.0 \\ 1 & 0 & 0 & 0 & 0.82 & 0 \end{bmatrix}$$

Config 3: $\theta = [0, 2.74, 1.0, 0, 0, 0]^T$

$$J_3 = \begin{bmatrix} 234.0 & -632.45 & 240.83 & 82.48 & -82.48 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 165.11 & 56.55 & -56.55 & 0 \\ 0 & 0 & 0 & 0 & 0.57 & 0 \\ 0 & -1 & 1 & -1 & 0 & -1.0 \\ 1 & 0 & 0 & 0 & -0.82 & 0 \end{bmatrix}$$

Config 4: $\theta = [0, 0, 1.0, 0, 0, 0]^T$

$$J_4 = \begin{bmatrix} 234.0 & 582.77 & -157.77 & -54.03 & 54.03 & 0 \\ 245.71 & 0 & 0 & 0 & 0 & 0 \\ 0 & 245.71 & -245.71 & -84.15 & 84.15 & 0 \\ 0 & 0 & 0 & 0 & -0.84 & 0 \\ 0 & -1 & 1 & -1 & 0 & -1.0 \\ 1 & 0 & 0 & 0 & 0.54 & 0 \end{bmatrix}$$

Config 5: $\theta = [0, -0.40, 0, 0, 0, 0]^T$

$$J_5 = \begin{bmatrix} 234.0 & 660.68 & -269.06 & -92.14 & 92.14 & 0 \\ -278.56 & 0 & 0 & 0 & 0 & 0 \\ 0 & -278.56 & 113.44 & 38.85 & -38.85 & 0 \\ 0 & 0 & 0 & 0 & 0.39 & 0 \\ 0 & -1 & 1 & -1 & 0 & -1.0 \\ 1 & 0 & 0 & 0 & 0.92 & 0 \end{bmatrix}$$

A. Jacobian Construction

The Jacobian matrix is constructed column by column, where each column represents the contribution of the corresponding joint to the end-effector velocity. For a planar manipulator, only the linear velocity components are considered, as the rotational rows are omitted for a 2-DOF system operating in the xy -plane. The Jacobian is partitioned into its linear and angular velocity components as

$$J_v = \begin{bmatrix} J_{x_v} \\ J_{y_v} \\ J_{z_v} \end{bmatrix}, \quad \text{and} \quad J_\omega = \begin{bmatrix} J_{x_\omega} \\ J_{y_\omega} \\ J_{z_\omega} \end{bmatrix}. \quad (26)$$

At joint 1, considering only the first joint's contribution, the linear velocity component is computed as

$$J_{v_1} = z_0^0 \times (o_1^0 - o_0^0), \quad (27)$$

giving the single-joint Jacobian

$$J_0^1 = \begin{bmatrix} J_{v_1}(1), 0 \\ J_{v_1}(2), 0 \end{bmatrix}. \quad (28)$$

At joint 2, the linear velocity contributions from both joints are computed as

$$J_{v_1} = z_0^0 \times (o_2^0 - o_0^0), \quad J_{v_2} = z_1^0 \times (o_2^0 - o_1^0), \quad (29)$$

where both joints are revolute. The full 2×2 Jacobian evaluated at the end-effector is then assembled as

$$J_0^2 = \begin{bmatrix} J_{v_1}(1) & J_{v_2}(1) \\ J_{v_1}(2) & J_{v_2}(2) \end{bmatrix}, \quad (30)$$

where $J_{v_i}(1)$ and $J_{v_i}(2)$ denote the x and y components of the i -th column respectively, and z_i^0 is the unit vector along the z -axis of frame i expressed in the base frame.

$$J_0^2 = \begin{bmatrix} -a_2 s_{12} - a_1 s_1 & -a_2 s_{12} \\ a_2 c_{12} + a_1 c_1 & a_2 c_{12} \end{bmatrix}, \quad (31)$$

where $s_1 = \sin(\theta_1)$, $c_1 = \cos(\theta_1)$, $s_{12} = \sin(\theta_1 + \theta_2)$, and $c_{12} = \cos(\theta_1 + \theta_2)$.

The following values, O_1 and O_2 , represent the positions of joints 1 and 2, respectively.

$$O_1 = \begin{bmatrix} \cos \theta_1 \\ \sin \theta_1 \end{bmatrix}, \quad (32)$$

$$O_2 = \begin{bmatrix} \cos \theta_1 + \theta_2 + \cos \theta_1 \\ \sin \theta_1 + \theta_2 + \sin \theta_1 \end{bmatrix}, \quad (33)$$

V. SIMULATION AND EXPERIMENTAL RESULTS

The Peter Corke Toolbox in MATLAB created the following simulation of each robotic arm configuration based on the D-H Convention in Table I and the theta values in Table III. Peter Corke's Robotics Toolbox in MATLAB was used to simulate each robotic arm configuration based on the D-H parameters in Table I and the joint angles listed in Table III.

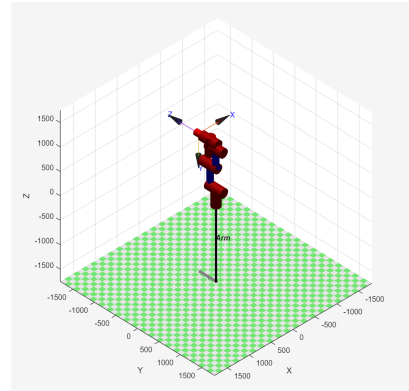


Fig. 2. Joint angles $\theta = [0, -90, 0, -90, 0, 0]$: The arm extends vertically upward from the base, with joints 2 and 4 folded at -90 producing a straight upright configuration that represents the robot's home position.

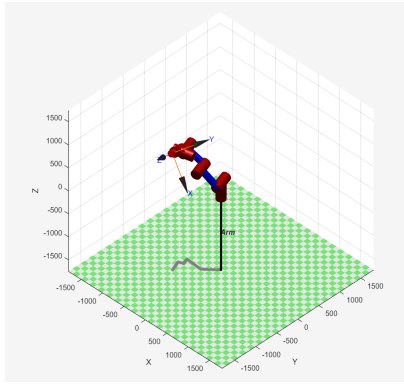


Fig. 3. Joint angles $\theta = [18, -43, 7.8, -60, 28, 40]$: The arm sweeps laterally with a moderate forward reach, with the elbow raised and the wrist angled downward, placing the end-effector at a mid-range position offset from the base.

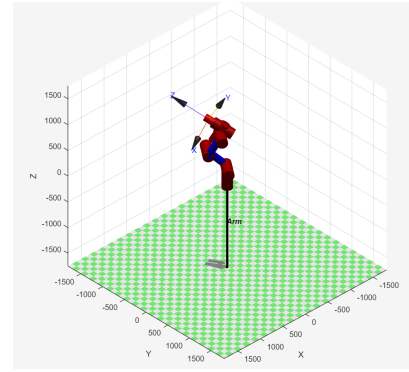


Fig. 6. Joint angles $\theta = [-36, -136, 77, -90, 70, 153]$: The arm sweeps to the opposite side of the base with the shoulder depressed steeply and the elbow bent forward, bringing the end-effector to a mid-height position with the wrist heavily rotated.

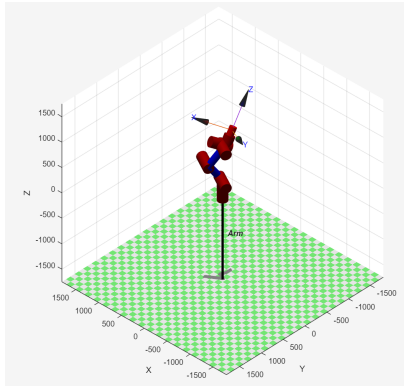


Fig. 4. Joint angles $\theta = [61, -125, 74, -11, 77, 80]$: The arm is configured with the shoulder pulled sharply backward and the elbow bent acutely, resulting in a compact, folded posture with the end-effector positioned low and close to the upper links.

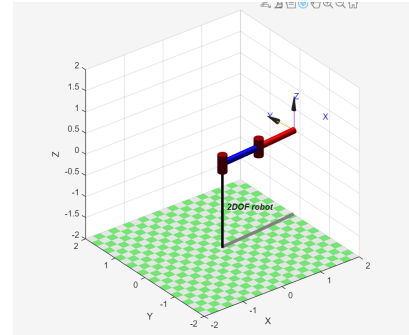


Fig. 7. The robot at the beginning of its simulation

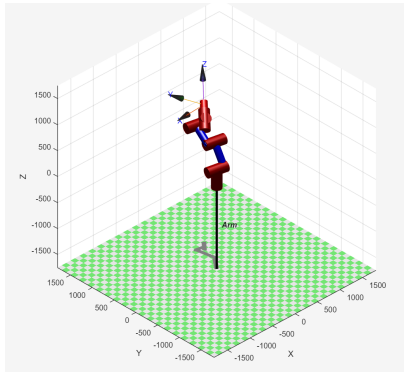


Fig. 5. Joint angles $\theta = [79, -130, -23, 8, 64, 108]$: The arm rotates significantly about the base joint and extends upward with the forearm nearly vertical, positioning the end-effector at a high elevation with considerable rotation in the wrist joints.

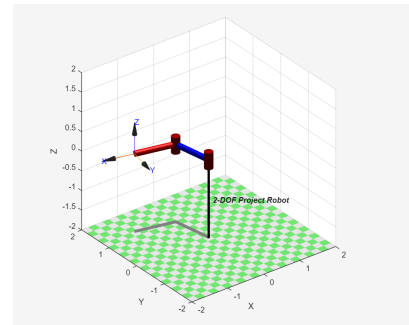


Fig. 8. The robot after the beginning of its simulation

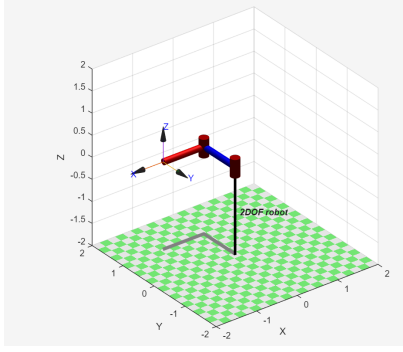


Fig. 9. The robot at the end of its simulation

The following three images represent the positions and angles of each joint per iteration, respectively. Both images represent the performance of the path planning system. Within Figure 10, it is shown that the joint positions do not take unnecessary movements towards the desired final configuration. In Figure 11, the joint angles quickly and accurately reach a stable value of $\frac{\pi}{2}$ at the desired final configuration. Figure 13 is a simplistic simulation of the robot arm, where each iteration is plotted together. The resulting graph shows that a minimal amount of iterations were necessary to achieve a desired final configuration.

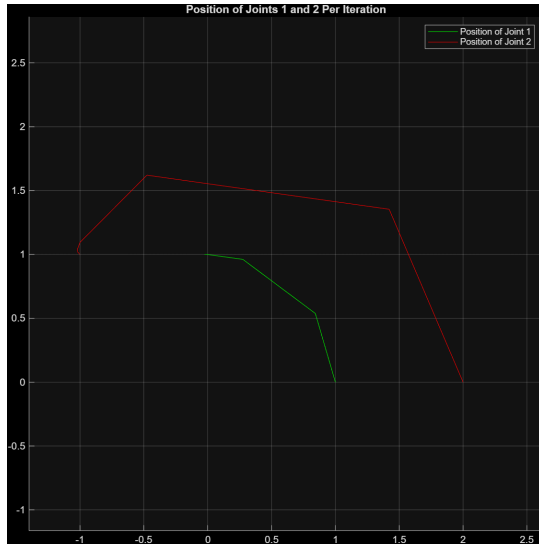


Fig. 10. Positions of the Joint Per Iteration

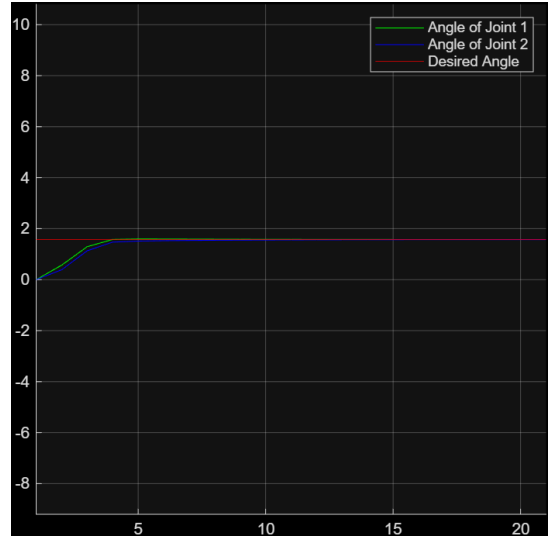


Fig. 11. Values of the Joint Angles Per Iteration

Fig. 12. Configurations per Iteration

VI. DISCUSSIONS

Several challenges were encountered throughout the course of this project. One of the most impactful errors involved incorrect input values for the joint variables θ_2 and θ_4 in the θ column of the D-H parameter table, which produced incorrect forward kinematics results until the issue was identified and corrected. Additionally, selecting appropriate gain values for the potential field path planning algorithm proved to be a time-consuming process, as the gains had to be tuned iteratively through trial and error until a trajectory that was smooth, fast, and overshoot-free was achieved. Finally, configuring and working with Peter Corke's Robotics Toolbox presented a significant learning curve, as getting the toolbox properly installed and producing correct robot animations required considerable time and troubleshooting before yielding usable simulation results.

VII. CONCLUSION

This project provided hands-on experience with the kinematic analysis and path planning of the UR5e robotic manipulator. The Denavit-Hartenberg convention was applied to derive the forward kinematics model of the robot from scratch, establishing the mathematical relationship between joint variables and the position and orientation of the end-effector relative to the base frame. The results of the forward kinematics model were validated against measurements taken from the physical UR5e robot using the teach pendant, with the remaining discrepancy attributed to the tool offset of the end-effector. The Jacobian matrix was then derived from the forward kinematics model, and five singular configurations were identified by solving for the joint configurations at which the determinant of the Jacobian vanishes. Finally, a potential field path planning algorithm was implemented for a 2-DOF

robot, successfully navigating the manipulator from its initial configuration to its final configuration while avoiding the obstacle. The resulting trajectory was smooth and overshoot-free, demonstrating the effectiveness of the potential field approach. Overall, this project reinforced the theoretical concepts covered in the course and developed practical skills in robotic modeling, simulation, and path planning using MATLAB.

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